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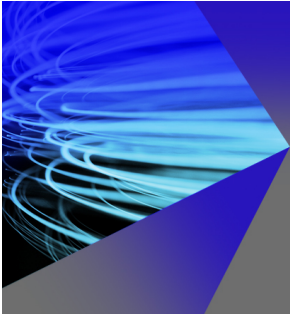
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ABSTRACT

Nickel/bismuth (Ni/Bi) bilayers have recently attracted attention due to the occurrence of time-reversal symmetry breaking in the superconducting state. Here, we report on the structural, magnetic, and electric characterization of thin film Ni/Bi bilayers with several Bi thicknesses. We observed the formation of a complex layered structure depending on the Bi thickness caused by the inter-diffusion of Bi and Ni which leads to the stabilization of NiBi₃ at the Bi/Ni interface. The superconducting transition temperature and the transition width are highly dependent on the Bi thickness and the layer structure. Magnetoelectric transport measurements in perpendicular and parallel magnetic fields were used to investigate the temperature-dependent upper critical field within the framework of the anisotropic Ginzburg–Landau theory and the Werthamer–Helfand–Hohenberg model. For thicker samples, we observed a conventional behavior, similar to that shown by NiBi₃ bulk samples, including a small Maki parameter ($\alpha_M = 0$), no spin–orbit scattering ($\lambda_{SO} = 0$) and nearly isotropic coherence length ($\gamma = \xi_{\perp}(0)/\xi_{\parallel}(0) \approx 1$). The values obtained for these properties are close to those characterizing NiBi₃ single crystals. On the other hand, in very thin samples, the Maki parameter increases to about $\alpha_M = 2.8$. In addition, the coherence length becomes anisotropic ($\gamma = 0.32$) and spin–orbit scattering ($\lambda_{SO} = 1.2$) must be taken into account. Our results unequivocally show that the properties characterizing the superconducting state in the Ni/Bi are strongly dependent on the sample thickness.

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I. INTRODUCTION

The bilayer formed by stacking bismuth (Bi) and nickel (Ni) presents a superconducting transition with a critical temperature (T_c) of approximately 4 K, in contrast to the absence of superconductivity (SC) in Ni and only below 0.5 mK for Bi.¹ This heterostructure was initially investigated by Moodera and Meserve² and LeClair *et al.*,³ using tunneling measurements to propose that SC potentially occurs across the entire bilayer. Subsequent experimental studies supported similar conclusions, providing evidence for a p -wave nature of symmetry pairing.^{4–6} Among the investigations presenting signatures of unconventional SC in the Ni/Bi bilayer, Gong *et al.* observed a notable spontaneous time-reversal symmetry breaking (TRSB) at the superconducting transition.⁷

Although many studies consider the Ni/Bi bilayer as a potential unconventional superconductor, the presence of a NiBi₃ layer formed at the interface may be an alternative explanation for the observed phenomenology. This particular intermetallic compound is described

as a strong coupling s -wave superconductor with T_c around 4.1 K,^{8–11} essentially the same T_c as for the Ni/Bi bilayer heterostructure. Regardless of the preparation technique, spontaneous formation of NiBi₃ is observed when the deposition temperature of the bilayer is higher than 110 K.^{12–17} However, when the Ni/Bi bilayer is prepared at 4.2 K, to avoid interdiffusion, Liu *et al.*¹⁴ verified that neither NiBi₃ is formed nor the resulting bilayer is superconducting.

In essence, the presence of an interfacial NiBi₃ layer appears to be a necessary condition for the observation of a superconducting transition in the Ni/Bi bilayer. A theoretical model proposed by Chao considers the presence of NiBi₃ acting as a mechanism to induce SC in the Bi layer via the proximity effect.¹⁸ In addition, the model takes into account the strong spin–orbit interaction related to Bi and the exchange coupling coming from Ni to provide a reasonable explanation for the p -wave-like Andreev reflection signatures seen previously.⁵ In other words, the model proposes a chiral p -wave symmetry with non-trivial topological phase to describe the

superconducting state in the Bi/Ni bilayer system. Therefore, at this point, it seems unclear whether the SC in Bi/Ni originates directly from NiBi_3 at the interface or whether it is induced by the proximity effect in the adjacent Bi layer. It is nonetheless unclear if these configurations host an unconventional superconducting state.

The study of these superconductors, possibly showing TRSB, is a fundamental step toward the understanding of the physics of unconventional superconductor, as well as, important for future technological advancements, since these superconductors have been considered as a great promise for quantum computing and spintronics.

To further contribute, we study the Bi/Ni bilayer system with its superconducting properties including: critical temperature, upper critical field, and coherence length as functions of the Bi layer thickness. Our analyses are based on Ginzburg–Landau and Werthamer–Helfand–Hohenberg (WHH) models, considering two different magnetic field configurations: parallel (aligned with the current) and perpendicular (out of the film plane). Our findings regarding the thickness dependence of the Bi layer in the Bi/Ni bilayer suggest the possibility of two distinct regimes within this system: one at higher thicknesses, where the superconducting properties are essentially ruled by the NiBi_3 bulk formed at the Bi/Ni interface; and another regime characterized by a thickness-dependent crossover point, at which unconventional properties may potentially manifest.

II. EXPERIMENTAL DETAILS

We grew Ni/Bi bilayer thin films on silicon oxide (100 nm) covered silicon wafer substrates by using the magnetron sputtering deposition technique. The substrates were kept at room temperature. We prepared several samples with a fixed 8 nm layer of Ni and layers of Bi with thicknesses ranging from 10 to 80 nm. The purity of both Ni and Bi targets was 99.99% and the power employed for the depositions was 65 W (RF) for the Ni target and 10 W (DC) for the Bi target. A capping Ti layer of 2 nm was applied to all samples to prevent oxidation. The vacuum chamber base pressure was maintained at 2×10^{-7} Torr, and the deposition working pressure was set to 1×10^{-3} Torr by using 200 SCCM flux of ultrapure argon gas (99.9999%).

Rutherford backscattering spectroscopy (RBS) using 1.5 MeV He^+ beam was employed to determine the samples' thicknesses. To study the crystallographic properties, we applied x-ray diffraction (XRD) in the $\Theta - 2\Theta$ configuration with $\text{Cu}\alpha_1$ radiation. Magnetization was measured as a function of the field at room temperature in a VSM EV9 magnetometer manufactured by Microsense Inc.

For the magneto-electrical measurements, we implemented a standard four-probe method where the in-plane current was applied at the extremities of the sample, while two inner contacts were used to measure the longitudinal voltage drop. In this arrangement, two magnetic field configurations were used: (i) parallel, with the field applied along the current direction, and (ii) perpendicular to the substrate plane. The magneto-transport experiments were performed using a setup based on a CMag9 cryo-cooler manufactured by Cryomagnetics Inc.

III. RESULTS AND DISCUSSION

A. Structural and magnetic characterization

Figure 1(a) displays the XRD patterns obtained for $\text{Ni}(8\text{ nm})/\text{Bi}(x)$ samples with $x = 10, 30$, and 60 nm . The presence of NiBi_3 peaks is evident in all samples, while Bi peaks occur only at $\text{Ni}(8\text{ nm})/\text{Bi}(60\text{ nm})$ sample. No peaks of individual Ni layer were observed. This result is in agreement with previous reports that also notice the formation of the inter-metallic NiBi_3 compound.^{12–14} The XRD pattern is consistent with the orthorhombic structure of NiBi_3 with $Pnma$ space group. We did not observe the NiBi compound in our samples, which contrasts with

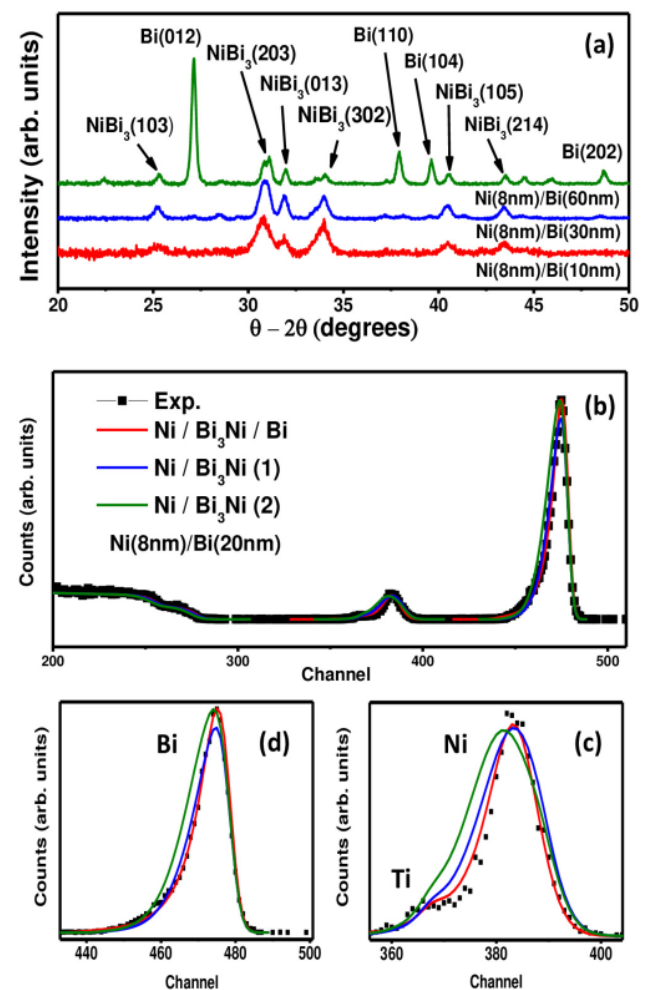


FIG. 1. (a) The XRD pattern of $\text{Ni}(8\text{ nm})/\text{Bi}(x)$, for $x = 10\text{ nm}$ (red), $x = 30\text{ nm}$ (blue), and $x = 60\text{ nm}$ (green). (b) The RBS spectra for $\text{Ni}(8\text{ nm})/\text{Bi}(20\text{ nm})$ with close-ups for the (c) Ni and Ti peaks and (d) Bi. The green and blue lines in panels (c) and (d) correspond to different SIMNRA simulations for a NiBi_3/Ni bilayer, while the red line is a simulation for a $\text{Ni}/\text{NiBi}_3/\text{Bi}$ trilayer.

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previous reports.^{14,19} The absence of double superconducting transitions in the RxT [see Fig. 3(a)] curve further supported this allegation. The lack of the BiNi phase could be related to the energies involved in deposition technique, as other works using sputter deposition also did not detect this phase.^{13,15,20}

Additional information on the structural properties was obtained from Gaussian fitting the XRD peaks in Fig. 1(a). The full-width at half maxima (FWHM) decreases with growing sample thickness, indicating that thicker samples are more crystalline. In the Ni(8 nm)/Bi(60 nm) sample (shown in green), the observed pure Bi peaks (012), (110), (104), and (202) are consistent with the R-3m trigonal structure. The presence of solid Bi suggests a limit in the formation of NiBi₃, which can be interpreted based on two contributions: (i) A diffusive limited process, in which the Bi diffuses into the Ni layer forming a Bi-Ni solid solution and then a NiBi₃ layer; or (ii) a diffusion-reaction limited process, in which Bi reacts with Ni forming the NiBi₃ layer until the available Bi or Ni being consumed.¹²

Figure 1(b) presents the RBS spectrum obtained for the Ni(8 nm)/Bi(20 nm) sample, whereas Figs. 1(c) and 1(d) provide zoomed-in views of the Ni and Bi peaks, respectively. The capping Ti layer is also detected as a shoulder in Fig. 1(d). To determine the sample stoichiometry and thickness, we utilized the SIMNRA software and the values of bulk mass density. In Figs. 1(c) and 1(d), the green and blue lines correspond to simulations of a Ni/NiBi₃ bilayer with: (1) focus on simulating the maxima of the Ni peak; and (2) focus on simulating the maxima of the Bi peak, whereas the red line corresponds to the simulation of the trilayer Ni/NiBi₃/Bi. The latter offers the best fit for the experimental data. This suggests that even for the bilayer with the lowest thickness, a thin layer of Bi remains after the formation of NiBi₃ at the interface. The depth profile extracted from the RBS spectrum is presented in Fig. S1 in the [supplementary material](#), and confirm the presence of three distinct layers in the sample. Additionally, the RBS simulation for the Ni(8 nm)/Bi(20 nm) indicates a thickness of 11 nm for Bi, 18 nm for NiBi₃ and 7 nm for Ni, approximately. This means that the relative consumption rate between Ni and Bi is close to (1:9) nm, which agrees with the findings of Vaughan *et al.*,¹³ who makes use of a polarized neutron scattering technique.

Figure 2 shows the magnetic moment normalized by the sample's area as a function of the in-plane applied magnetic field. These experiments were carried out at room temperature. Previous studies have not reported any magnetic order in NiBi₃,^{9,21} although inclusions of ferromagnetic Ni impurities have been observed in this alloy.²⁰ In any case, we assume that the magnetic moment observed in our samples originates entirely from the Ni layer. Results in Fig. 2 indicate that the presence of the Bi layer with increasing thickness gradually suppresses the Ni magnetic moment. The inset in Fig. 2 depicts the magnetic moment of saturation (M.S.) for different Bi thicknesses. Clearly, the M.S. drastically reduces for samples where the Bi layers have a thickness larger than 40 nm. The suppression of the M.S. is related to the consumption of the Ni layer to form the NiBi₃ compound at the interface. The Bi atom can diffuse throughout the pre-formed NiBi₃ layer reaching to Ni atoms, reacting and forming a new layer of NiBi₃, thus, reducing the amount of metallic/magnetic Ni volume. This

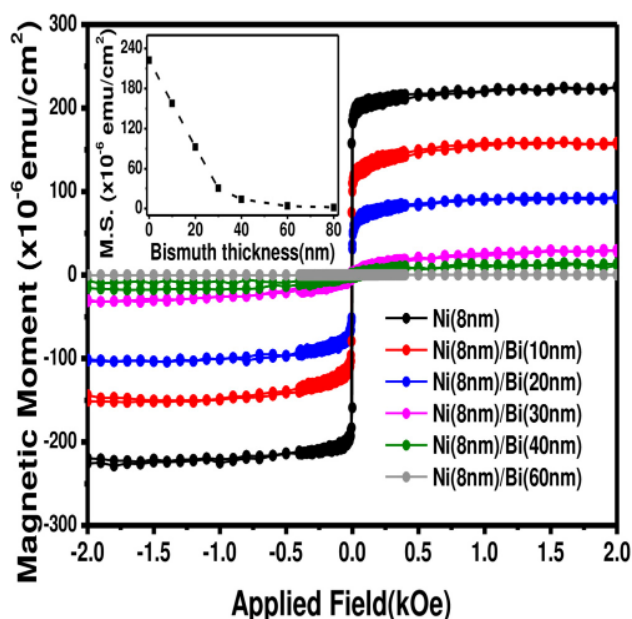


FIG. 2. Magnetic moment normalized by the area of each sample. Measurements were made at room temperature as functions of the in-plane magnetic field. Inset: magnetic moment of saturation (M.S.) normalized by area vs bismuth thickness.

interpretation claims in favor of the diffusion-reaction limit process previously discussed by Siva *et al.*¹²

We estimated the thickness of the ferromagnetic portion of the Ni layer by assuming the bulk magnetization and structural parameters. Interestingly, for an 8 nm pure Ni layer (black dots in Fig. 2) the estimation yielded an effective ferromagnetic Ni layer thickness of about 4.8 nm. This suggests the occurrence of magnetically dead Ni sub-layers within the thin film structure. Such an interpretation was previously proposed by Liebermann *et al.* to account for similar observations in Ni thin films at room temperature.²² Regarding the Ni/Bi bilayers, we estimate thicknesses for the ferromagnetic Ni of 3.4 nm, 2.1 nm, and 0.8 nm for Ni(8 nm)/Bi(*x*) with *x* = 10, 20, and 30 nm, respectively. These results indicate a consistent consumption rate close to (1:9) nm, similar to the one obtained from the RBS analysis.

Based on characterization, we are led to the conclusion that two distinct types of Ni/Bi layered structures are present in our bilayers. For Bi depositions up to 30 nm, we effectively obtained a Ni/NiBi₃/Bi trilayer, resembling a heterostructure configuration of metal/superconductor/ferromagnet. In contrast, Bi thicknesses exceeding 30 nm reveal the occurrence of a progressive suppression of ferromagnetism in the Ni film, so that the samples behave as a NiBi₃/Bi bilayer, which is akin to a metal/superconductor system. The change in the sample structure arises from the reaction-diffusion mechanism, facilitating the diffusion of Bi atoms from the upper layer to react with Ni atoms until the Ni layer is fully consumed. In our scenario, this process requires approximately 30 nm

of Bi. A similar result was observed by Siva *et al.*,¹⁷ demonstrating that around 7 nm of the Ni layer was consumed by a 30 nm Bi layer (in this case two separate layers of 15 nm). Here, we consider that the Bi layer behaves as a metallic component, which is expected due to its low thickness.²³

B. Thickness-dependence of the critical temperature

Figure 3(a) shows the normalized resistance (R/R_{5K}) vs temperature, while Fig. 3(b) displays T_c as a function of the Bi thickness. Previous reports showed that a superconducting transition around 6 K could appear in Bi thin films due to an amorphous

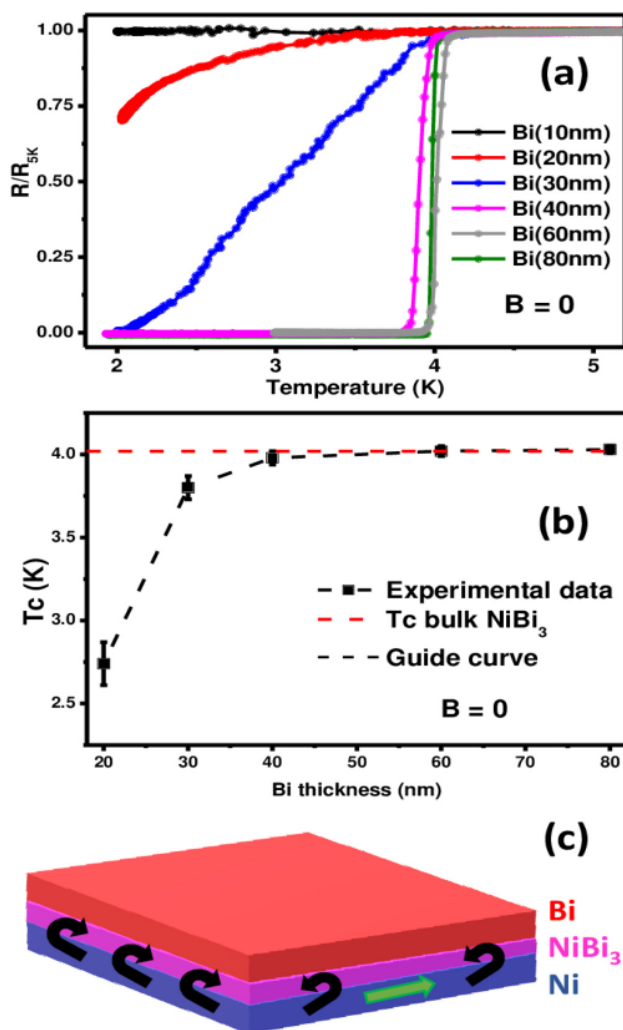


FIG. 3. (a) Normalized resistance as a function of temperature for Ni(8 nm)/Bi(x) samples. (b) T_c as a function of Bi thickness. (c) Schematic illustration of the Ni/Bi₃Ni/Bi trilayer. The black arrows indicate the stray field originating from the Ni layer and green arrow corresponds to the in-plane magnetization.

crystalline phase,¹⁴ however no transition was observed close to this temperature in our samples (see Fig. S2 in the [supplementary material](#)). We notice that the RxT behavior of the Bi/Ni samples, displayed in the [supplementary material](#) at Fig. S2, is typical of the NiBi_3 system.^{9–11,20} Additionally, no sign of resistance increment was observed at low temperatures, as observed in Ref. 16. Since our lowest reachable temperature is 1.8 K, we define T_c as the temperature where the measured resistance decreases to 90% of the normal state resistance. Similar profiles of T_c vs thickness were reported in Refs. 5, 7, 13, 14, and 15 for Ni/Bi bilayers. For samples deposited with higher Bi thickness, i.e., Ni(8 nm)/Bi(40 nm), Ni(8 nm)/Bi(60 nm), and Ni(8 nm)/Bi(80 nm), the superconducting transition is sharp and has a $T_c \approx 4.05$ K, similar to those reported for the NiBi_3 compound [red dashed-line in Fig. 3(b)]. On the other hand, a large broadening in the superconducting transition is observed for the samples Ni(8 nm)/Bi(20 nm) and Ni(8 nm)/Bi(30 nm). The transition for the sample Ni(8 nm)/Bi(10 nm) was not observed to occur in the measured temperature range. The broadening of the superconducting transition is usually related to the disorder. We have indeed observed some increase in lattice disorder by looking at the FWHM in the XRD patterns for samples with small Bi thicknesses. Nonetheless, we would like to point out that the transition widens quite significantly when the Bi thickness is reduced below the 40 nm threshold. For instance, the transition broadening in the Ni(8 nm)/Bi(30 nm) sample is about 10 times larger than that in the Ni(8 nm)/Bi(40 nm) specimen. The FWHM, however, changes much more slowly in the same range of Bi thickness. Therefore, we believe that the broad transition might be related to other effects apart from lattice disorder. By comparing the T_c curve in Fig. 3(b) with the inset in Fig. 2, we observe that a strong widening of the superconducting transition occurs in samples with a non-zero magnetic moment. This leads us to speculate that the remaining ferromagnetic Ni layer generates a highly non-uniform magnetic stray field, as depicted in Fig. 3(c). This stray field acts on the NiBi_3 layer and causes a significant detrimental effect on the superconducting state of the bilayers, particularly for the samples with thinner superconducting layers, resulting in the sharp reduction of T_c . The influence of stray fields in Ni/Bi bilayers has been discussed in a previous study.²⁴

The primary factor contributing to the broadening of the red and blue transition curves in Fig. 3(a) probably comes from the stray field generated by the nearby ferromagnetic Ni. Although less likely, there might be another contribution to this effect: a crossover from a three-dimensional (3D) to two-dimensional (2D) superconductivity. This is expected to occur when the thickness of the superconducting layer becomes comparable to or smaller than the coherent length. The change in dimensionality induces significant fluctuations in the order parameter due to the influence of interface and surface scattering, resulting in a strong decrease of T_c , similar to that shown in Fig. 3(b), and changes on the upper critical field vs T phase diagram below a specific temperature.^{25–28} We will return to this point later in this manuscript.

C. Superconducting properties

In Fig. 4 we show the normalized resistance vs temperature results measured at different magnetic fields for two field

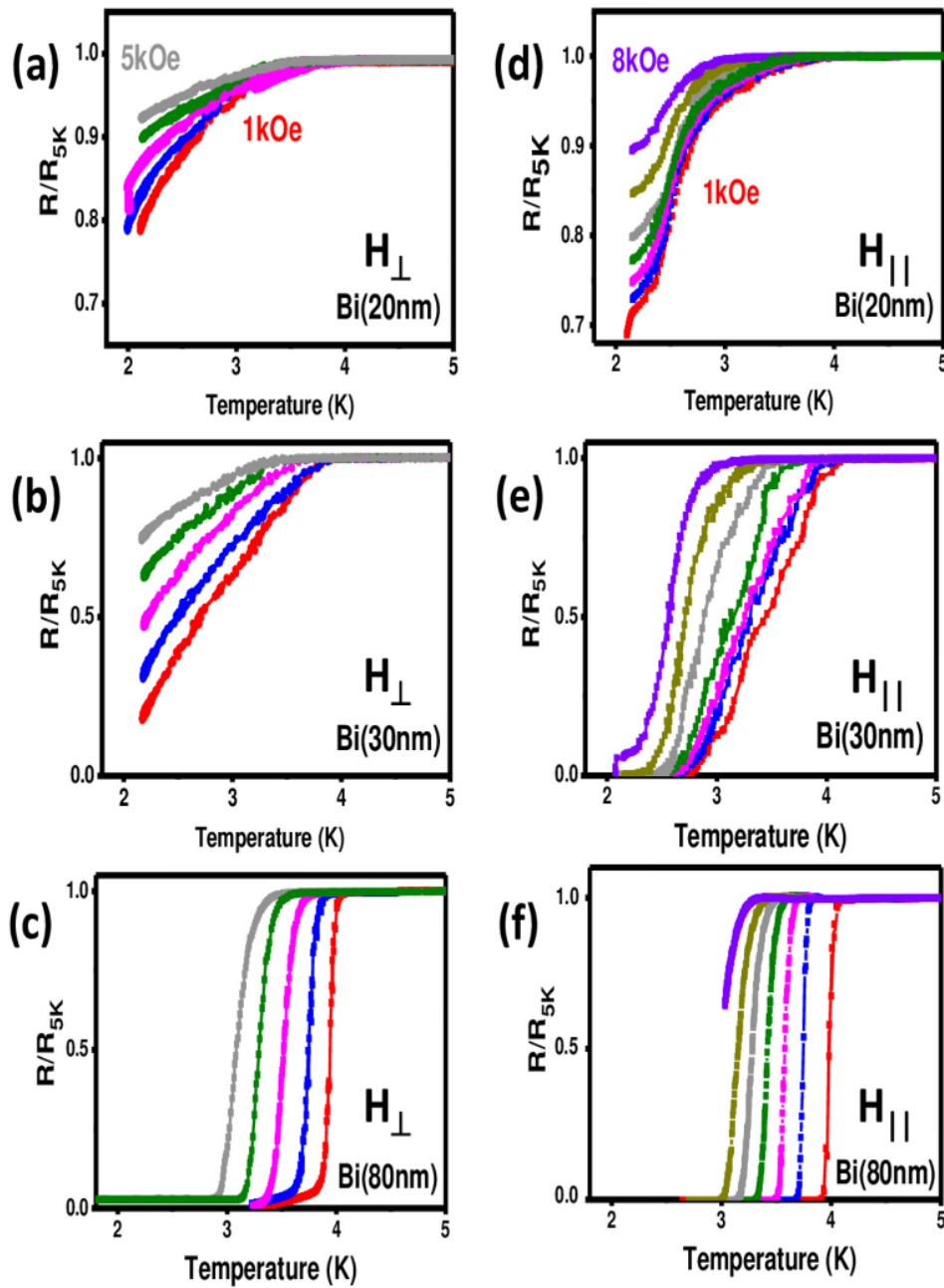


FIG. 4. Normalized resistance in 5 K vs temperature for perpendicular (a)–(c) and parallel (d)–(f) applied magnetic field orientation for Ni(8 nm)/Bi(x) samples with $x = 20, 30$, and 80 nm. The colors start from 1 kOe (red) up to 5 kOe (gray) for perpendicular and up to 8 kOe (purple) for parallel magnetic fields.

orientations, namely, perpendicular to the plane of the film (a)–(c) and parallel (d)–(f) to the electrical current. Applied fields are from 1 to 5 kOe in steps of one (colors varying from red to gray), and 1 to 8 kOe (colors varying from red to purple) for parallel and

perpendicular field directions, respectively. Corrections for the demagnetizing effects are negligible in all cases. Increasing fields move the transition $T_c(B)$ toward lower temperatures as expected. Nevertheless, the width of the transition seems to increase much

faster in samples with low Bi thickness, which can also be related to the stray field generated by the ferromagnetic Ni layer.^{29,30} From Fig. 4, we extract the upper critical field $B_{c2,\perp}$ and $B_{c2,\parallel}$ as a function of the reduced temperature $t = T/T_c$. The obtained experimental points are shown in Fig. 5.

In the case of perpendicular magnetic fields, the transition from the normal state to the superconducting vortex phase is described by the linear temperature dependence shown by the red points in Figs. 5(a)–5(c) for all samples. Assuming that the orbital effect is the dominant limiting mechanism introduced by the applied magnetic

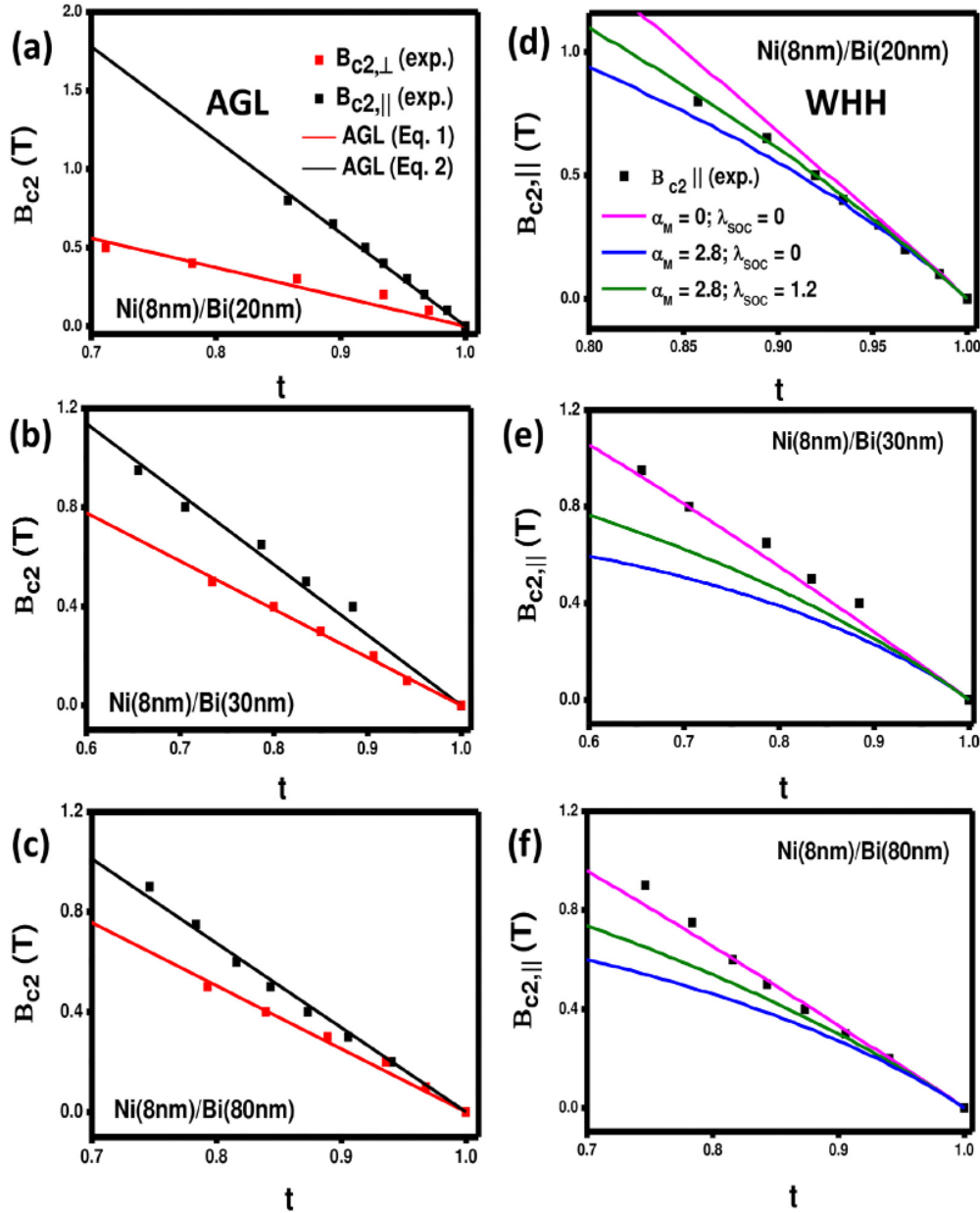


FIG. 5. Upper critical field vs temperature phase diagram analyzed within the scope of AGL theory in perpendicular [Eq. (1)] and parallel 3D [Eq. (2)] fields for Ni(8 nm)/Bi(x) (x) (a) $x = 20$ nm; (b) $x = 30$ nm and (c) $x = 80$ nm. Parallel upper critical field in function of reduced temperature within the scope of WHH model for Ni(8 nm)/Bi(x) (d) $x = 20$ nm; (e) $x = 30$ nm and (f) $x = 80$ nm.

field, we estimate the in-plane zero-temperature coherence length $\xi_{\parallel}(0)$ by fitting the red experimental data to the following expression from the anisotropic Ginzburg–Landau (AGL) model,²⁶

$$B_{c2,\perp}(t) = \frac{\Phi_0}{2\pi\xi_{\parallel}^2(0)}(1-t), \quad (1)$$

where Φ_0 is the flux quantum. The extracted values for $\xi_{\parallel}(0)$ are listed in Table I. The values of $\xi_{\parallel}(0)$ are close to those reported by Vaughn¹³ for a Ni/Bi bilayer of about 13.8 nm and slightly lower than those for a NiBi₃ single crystal,⁹ where $\xi_a(0) = 18.1$ nm and $\xi_c(0) = 14.3$ nm along crystalline *a* and *c* axis, respectively. Another relevant result is the weak dependence of $\xi_{\parallel}(0)$ with respect to the Bi thickness. Here, one would expect a reduction of $\xi_{\parallel}(0)$ (bulk-like) with the reduction of thickness,³¹ once thinner samples should present higher levels of disorder, as indeed shown by the residual resistivity ratio (RRR) also displayed in Table I. Consequently, we are led to consider that disorder does not play a significant role in the properties of the superconducting state of our samples, at least those are directly dependent on the coherence length.

The experimental data corresponding to in-plane magnetic fields are represented by the black points in Figs. 5(a)–5(c). We fitted these results to the 3D AGL model,²⁶ which predicts that the upper critical field for in-plane direction is given by

$$B_{c2,\parallel} = \frac{1.69\Phi_0}{2\pi\xi_{\parallel}(0)\xi_{\perp}(0)}(1-t), \quad (2)$$

where the out-of-plane coherence length ($\xi_{\perp}$) is introduced by assuming the effective mass anisotropy approximation. By fitting the in-plane field data with the above expression [black line on Figs. 5(a)–5(c)] we extracted the values of $\xi_{\perp}(0)$, which are also listed in Table I.

For samples with 30 and 80 nm of Bi, the anisotropy factor defined as $\gamma = \xi_{\perp}(0)/\xi_{\parallel}(0)$ is close to 1, indicating an isotropic behavior similar to the bulk NiBi₃ type-II superconductor. On the other hand for the Ni(8 nm)/Bi(20 nm) specimen, the $\xi_{\perp}(0)$ is smaller than $\xi_{\parallel}(0)$, revealing the occurrence of an anisotropy in the coherence length that is characterized by the $\gamma = 0.3$. At first glance, one might consider that the reduction in $\xi_{\perp}(0)$ would be related to 2D effects due to the reduction of the NiBi₃ thickness. However, the 2D GL theory predicts that $B_{c2,\parallel}$ varies with temperature as $\sim (1-t)^{1/2}$,³² which does not fit to our data.

Another possibility is the effect of a dimensional crossover. The enhancement of $B_{c2,\parallel}$ provoked by the 3D–2D dimensional crossover usually occurs below a characteristic temperature known as T^* .^{25–28} In our case, however, the linear temperature dependence

shown by the experimental data clearly implies the validity of the $T^* < T < T_c$ regime. For this reason, the parallel upper critical field should approach the 3D value of the ordinary type-II superconducting phase. Based on these arguments, we speculate that the reduction of $\xi_{\perp}(0)$ in thin samples may be related to an effect other than disorder and dimensional crossover.

The Ni/Bi system has been widely investigated due to the possibility of hosting unconventional superconductivity, where the spin–orbit coupling is supposed to play a key role. Assuming that the superconductivity in our bilayers stems from the NiBi₃ layer, this inevitably leads to a superconducting state that has a different environment close to the interfaces with Bi and Ni. We then speculate that the system studied here, the trilayer Ni/NiBi₃/Bi, in the limit of low Bi thickness, as we observed in the sample Ni(8 nm)/Bi(20 nm), could belong to the class of noncentrosymmetric superconductors. These materials experience inversion-breaking spin–orbit coupling effects enhancing the critical field.

To investigate the role of spin–orbit coupling in this system, we analyze the $B_{c2,\parallel}$ with the Werthamer–Helfand–Hohenberg (WHH) model. Three calculated theoretical curves and the experimental points are shown in Figs. 5(d)–5(f). This model incorporates both orbital and paramagnetic limiting mechanisms. The relative strength between these two effects is quantified by the Maki parameter,³³ defined as $\alpha_M = \sqrt{2}B_{c2,\text{orb}}(0)/B_p(0)$, where $B_{c2,\text{orb}}$ and B_p are the orbital and paramagnetic limiting critical fields at zero temperature, respectively. For a single-band dirty-limit superconductor, the WHH model includes α_M and spin–orbit scattering (λ_{SO}). The overall upper critical field is obtained from the equation below,³⁴

$$\ln\left(\frac{1}{t}\right) = \sum_{v=-\infty}^{\infty} \left(\frac{1}{|2v+1|} - \left[|2v+1| + \frac{\bar{h}}{t} + \frac{(\alpha_M \bar{h}/t)^2}{|2v+1| + (\bar{h} + \lambda_{SO})/t} \right] \right),$$

$$\text{where } \bar{h} = \left(\frac{4}{\pi^2} \frac{B_{c2}(t)}{dB_{c2}(t)/dt} \right)_{t=1}.$$

Since the AGL model includes purely orbital effects, we considered $B_{c2,\text{orb}}(0)$ as given by the slope obtained from fits with Eq. (2) near T_c , while $B_p(0) = 1.84T_c$ is established in the literature for BCS superconductors.³⁵ Thus, for Ni(8 nm)/Bi(20 nm), Ni(8 nm)/Bi(30 nm) and Ni(8 nm)/Bi(80 nm) samples we roughly estimate α_M as 2.80, 0.34, and 0.38, respectively.

Figures 5(e) and 5(f) present the WHH fit for the thicker samples, with Bi layers of 30 and 80 nm. Notice that the best agreement to the experiments occurs for both parameters α_M and λ_{SO} equal to zero (magenta curve). This situation corresponds to $B_{c2,\text{orb}}(0) \ll B_p(0)$ indicating a field-induced suppression of superconductivity due to orbital effects, as expected for ordinary type-II superconductors. The case of Ni(8 nm)/Bi(20 nm) sample is more interesting, once the experimental points can be well described by introducing $\alpha_M = 2.8$ and $\lambda_{SO} = 1.2$, see Fig. 5(d). It is essential to emphasize that our plots do not guarantee the existence of spin–orbit coupling. However, it can be considered as evidence that, at low thicknesses, the manifestation of spin–orbit effects can potentially account for the changes observed in the superconducting properties of the Ni/Bi system.

The origin of SC in the Ni/Bi bilayers is still a topic of debate. Whether it occurs at the Bi layer near the interface or whether it

TABLE I. Extracted coherence lengths within the framework of anisotropic Ginzburg–Landau theory, anisotropy factor γ and residual resistivity ratio— $R_{300\text{ K}}/R_{5\text{ K}}$.

Sample	$\xi_{\parallel}(0)$ (nm)	$\xi_{\perp}(0)$ (nm)	$\gamma = \xi_{\perp}/\xi_{\parallel}$	RRR
Ni(8 nm)/Bi(20 nm)	13.3 ± 0.4	4.2 ± 0.4	0.3	1.7
Ni(8 nm)/Bi(30 nm)	13.1 ± 0.2	15.1 ± 0.6	1.2	1.8
Ni(8 nm)/Bi(80 nm)	11.1 ± 0.3	14.5 ± 0.3	1.3	3.1

stems from bulk NiBi_3 formed at the interface remains a controversial subject. In this work, we find evidence in favor of the second hypothesis for thicker samples. In contrast, we point out the possibility of interesting effects occurring when very thin layers of NiBi_3 are formed at the bilayer interface. In this limit, the lack of inversion symmetry and the low thickness of the interfacial superconducting compound leads to a peculiar phenomenology such as that originated by an anti-symmetric spin-orbit coupling.³⁶ This adds to the claims favoring unconventional superconductivity reported so far in the Ni/Bi system^{4–7} and gives support to theoretical models as the one in Ref. 18. Additionally, we notice that bulk NiBi_3 has been recently suggested as hosting a topological surface state,³⁷ which might be a potential ingredient to be considered in the physics of thin Ni/Bi bilayers.

IV. CONCLUSIONS

We investigated the structural properties and the temperature dependence of the upper critical field of polycrystalline $\text{Ni}(8\text{ nm})/\text{Bi}(x)$ bilayers with $x = 10, 20, 30, 40, 60$, and 80 nm . XRD, RBS, and VSM analysis indicated that the original bilayers actually form the trilayer structure $\text{Ni}/\text{NiBi}_3/\text{Bi}$. For samples with large Bi thicknesses, the structure becomes effectively NiBi_3/Bi . In the trilayer, the stray field due to the remaining layer of ferromagnetic Ni leads to a decrease in the critical temperature and to a broadening of the superconducting transition. The upper critical field was analyzed in the context of the AGI and WHH models. For larger Bi layer thicknesses, the coherence length is nearly isotropic ($\gamma = \xi_{\perp}(0)/\xi_{\parallel}(0) \approx 1$) and has values similar to that found for NiBi_3 single crystals. On the other hand, for low Bi thicknesses, an anisotropic behavior is observed ($\gamma = 0.32$). In the framework of the WHH model, thicker samples are described by a null Maki parameter and spin-orbit coupling. In the smallest Bi thickness sample, the Maki parameter ($\alpha_M = 2.8$) and the spin-orbit scattering ($\lambda_{SO} = 1.2$) are shown to be relevant to fully describe the temperature dependence of the upper critical field. Therefore, our results favor a thickness-dependent superconducting state in the Ni/Bi bilayer system.

SUPPLEMENTARY MATERIAL

See the supplementary material for some additional information on RBS, especially the depth profile analysis. Also some data about the R vs T curve ranging from 300 to 2 K are included, along with a comparative normalized electrical curve for different samples spanning from the 120 to 2 K temperature range.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Gabriel Sant'ana da Silva Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **David Möckli**: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **A. D. C. Viegas** Resources (equal); Writing – review & editing (equal). **Paulo Pureur**: Formal analysis (equal); Funding acquisition (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Milton A. Tumelero**: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Resources (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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